
Axiomatic System Viability Model (ASVM): A Unified Framework for Dynamic Constraint Optimization in Complex Adaptive Systems

Abstract: The inherent complexity of modern systemic challenges necessitates a formal shift from qualitative philosophical description to quantitative, executable axiomatic modeling. This paper proposes the Axiomatic System Viability Model (ASVM), an optimal control framework designed to assess and stabilize complex adaptive systems ($\mathbf{x}(t)$). ASVM unifies principles from Viability Theory, Optimal Control Theory, and Fuzzy Set Logic to define system coherence through a continuous objective functional $\mathbf{J}$. Critically, the model formalizes meta-axiomatic shifts—such as the loss of feasible constraints (State P_2) or high internal conflict (P_1)—into actionable optimization queries ($\mathbf{X}^*$), ensuring that systemic resilience is achieved by dynamically redefining the system’s ideal goal state.

1. Introduction and Foundational Axioms

Complex adaptive systems operate under a set of non-negotiable constraints (axioms) that define their operational space. When these axioms are challenged—whether through internal contradiction or external shock—the system must undergo a structured, mathematically governed process of adaptation rather than collapse. The ASVM posits that systemic viability is not merely the maximization of utility, but the continuous maintenance of feasibility within a dynamically changing constraint set.

1.1. System Dynamics and State Space

The state of the system $\mathbf{x}(t) \in \mathbb{R}^n$ evolves according to a deterministic yet controlled differential equation:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}(t)) + \mathbf{u}(t)$$

where $f(\mathbf{x}(t))$ represents the intrinsic, unmanaged dynamics of the system (the natural drift), and $\mathbf{u}(t)$ is the aggregated control input vector:

$$\mathbf{u}(t) = \begin{bmatrix} u_{\mathcal{S}}(t) \\ u_{\mathcal{H}}(t) \\ u_{\mathcal{K}}(t) \end{bmatrix}$$

The components u_S , u_H , and u_K represent actionable interventions related to **Structure** ($\mathcal{S}$), **Growth/Hypothesis** ($\mathcal{H}$), and **Decay/Maintenance** ($\mathcal{K}$), respectively. These inputs are constrained by a time-varying feasible set $\Omega(t)$.

1.2. Foundational Theoretical Convergence

The ASVM is mathematically anchored in three distinct fields:

1. **Viability Theory (Aubin, 1991):** Defines the continuous constraints $g(\mathbf{x}(t)) \leq 0$. System stability requires that a solution path exists *within* the viable domain:

$$\Omega(t) = \{\mathbf{x}(t) \in \mathbb{R}^n \mid g(\mathbf{x}(t)) \leq 0\}$$

2. **Optimal Control Theory (Pontryagin et al., 1962):** Provides the mechanism for maximizing system performance ($\mathbf{J}$) by selecting optimal control inputs $\mathbf{u}^*(t)$ subject to the dynamic system constraints.
3. **Fuzzy Set Logic (Zadeh, 1965):** Manages inherent contradictions within the axioms, providing a quantifiable measure of axiomatic failure when binary truth values are insufficient.

2. The Core Optimization Problem: Defining Systemic Trajectory

The primary goal is to find the optimal control trajectory $\mathbf{u}^*(t)$ that maximizes performance while rigorously respecting all systemic constraints. This is framed as an optimal control problem:

$$\max_{\mathbf{u}(t) \in \Omega(t)} \mathbf{J}[\mathbf{u}] = \int_0^T \mathcal{L}(\mathbf{x}(t), \mathbf{u}(t), t) dt$$

where $\mathcal{L}(\cdot)$ is the Lagrangian functional representing immediate performance yield minus structural cost.

2.1. The Conflict Metric and State P_1 (Axiomatic Contradiction)

When the system faces a conflict between its Veracity ($\mu_{\mathcal{T}}$ —the factual consistency of knowledge) and its Utility ($\mu_{\mathcal{E}}$ —the immediate functional usefulness), this disagreement must trigger an intervention.

The Conflict Measure $C(I)$ quantifies the weighted dissonance between these two fuzzy membership values:

$$C(I) = |w_T \cdot \mu_{\mathcal{T}}(I) - w_E \cdot \mu_{\mathcal{E}}(I)|$$

where $w_T, w_E \geq 0$ (such that $w_T + w_E = 1$) are empirically calibrated weighting factors defining current axiomatic priority.

- **Trigger Condition:** If $C(I)$ exceeds a critical threshold ϵ , the system enters **State P_1 (Axiomatic Conflict)**, necessitating an immediate restructuring of the objective function $\mathbf{J} \rightarrow \mathbf{J}'$.

2.2. The Empty Feasible Set and State P_2 (Structural Failure)

The most severe failure mode occurs when constraints become mutually contradictory, rendering no operational state possible. This is formally defined as the empty feasible set:

$$\text{State } P_2 \text{ Condition: } \Omega(t) = \emptyset$$

When $\Omega(t) = \emptyset$, the standard maximization problem becomes ill-posed. The system must execute a **Meta-Axiomatic Restructuring (The “Molt”)** to redefine the fundamental constraints $g(\mathbf{x})$ and re-establish a viable domain $\Omega'(t) \neq \emptyset$.

3. Solving for Systemic Coherence: The Solution Seed ($\mathbf{X}^*$)

When $\mathbf{J}$ is rendered unachievable due to $\Omega(t) = \emptyset$, the ASVM employs an inverse optimization search to determine the minimal boundary perturbation required to restore viability. This optimal intervention parameter is the **Solution Seed ($\mathbf{X}^*$)**.

The Solution Seed $\mathbf{X}^*$ is found by solving a constrained optimization problem evaluated along the boundary of the infeasible region ($\partial\Omega$):

$$\mathbf{X}^* = \arg \min_{\mathbf{X} \in \partial\Omega} \|\mathbf{J}(\mathbf{x} | \mathbf{X}) - \mathbf{J}_{\text{target}}\|^2$$

The selection mechanism for $\mathbf{X}^*$ is constrained by **parsimony**: it selects the parameter adjustment requiring the minimal Euclidean distance in constraint space while satisfying feasibility.

4. Conclusion and Future Work

The Axiomatic System Viability Model provides a rigorous, unified mathematical framework for analyzing systemic coherence under stress. By translating theoretical concepts like “contradiction” (P_1) and “impossibility” (P_2) into quantifiable optimal control problems, ASVM achieves a level of academic rigor suitable for interdisciplinary peer review.

Future work will focus on developing real-time computational classes (e.g., approximate dynamic programming or deep reinforcement learning heuristics) necessary to compute the high-dimensional $\mathbf{X}^*$ query in operational timescales.

References

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